BURMESE PRAGMATICS

NLP CLASSIFICATION SYSTEM

A 4-STAGE ABLATION STUDY ON CONTEXT-SENSITIVE PRAGMATIC CLASSIFICATION IN BURMESE

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THE PROBLEM AND HOW WE GOT HERE

WHAT WE ARE CLASSIFYING

Given a Burmese utterance, predict its politeness level — one of 6 classes:

01 neutral
02 polite
03 informal
04 professional
05 blunt
06 rude

Burmese is a low-resource language where the same idea is expressed very differently depending on who is speaking, to whom, and in what context. This is called pragmatics.

ORIGINALLY: GENERATION

- Model: MT5 (encoder-decoder)
- Task: generate the Burmese utterance from context
- Training loss ~27, never converged
- One run: loss=0.0, val=NaN (GPU/FP16 bug)
- BLEU score meaningless — 100 outputs all valid for same input
- Conclusion: architecturally wrong

NOW: CLASSIFICATION

- Model: xlm-roberta-base (encoder only)
- Trained on 100 languages including Burmese
- Task: predict politeness label
- Metrics: Accuracy + Macro F1
- Macro F1 treats all 6 classes equally — critical for imbalanced data
- Result: works, improves with more context

DATASET & PREPROCESSING

2,200	22	~100	CC0
UTTERANCES	ROOT MEANINGS	VARIATIONS EACH	PUBLIC DOMAIN

DATASET: FREOCOCO/BURMESE-CONTEXTUAL-PRAGMATICS

POLITENESS: 13 → 6 CLASSES

neutral, polite, informal,	keep as-is
professional, blunt, rude	keep as-is
very_polite, religious,	→ polite
polite_but_firm	→ polite
friendly	→ informal
impolite, sarcastic	→ rude

CLASS IMBALANCE FIX

- neutral = 53% of data
- Without fixing: model predicts neutral for everything → 53% accuracy but learns nothing
- Fix: weighted cross-entropy loss — rare classes get higher weight
- Evaluation: Macro F1 — treats all 6 classes equally regardless of size

TONE + SPLIT

- 131 raw tone values:
- informal annotations, typos, compound labels — unlearnable
- Mapped to 5 categories via Appraisal Theory (Martin & White, 2005):
- neutral · positive · formal · negative · emotional

- Split: 70/15/15 train/val/test
- Seed=42, identical across all 7 models
- No data leakage

EXPERIMENTAL DESIGN: 4-STAGE ABLATION

STAGE 1
UTTERANCE ONLY

INPUT:

UTTERANCE

Baseline — can XLM-R read Burmese?

STAGE 2
+ CONTEXT

INPUT:

UTTERANCE + CONTEXT + INSTRUCTION

Does the social situation help?

STAGE 3
ORACLE METADATA

INPUT:

[REGISTER] [POWER] [TONE] + UTTERANCE + CONTEXT

Upper bound with perfect metadata

STAGE 4
PREDICTED METADATA

INPUT:

SAME AS STAGE 3 BUT METADATA PREDICTED BY CLASSIFIERS A/B/C

Real end-to-end pipeline

INTERMEDIATE CLASSIFIERS — A, B, C

Three auxiliary classifiers predict pragmatic metadata for Stage 4.
Each is independently fine-tuned from xlm-roberta-base on the same train/val/test split.

Classifier A — Register

Input: Utterance only

Classes: colloquial · standard · formal · slang

Why: Register is in the text — vocabulary and particles

Macro F1: 0.70

Hardest: standard (0.53 F1)
Standard sits ambiguously between colloquial and formal — inherently hard

Classifier B — Power Relation

Input: Utterance + context

Classes: equal · inferior_to_sup · superior_to_inf · any (merged 8→4)

Why: Power relation is social — needs situational context

Macro F1: 0.60

Hardest: any (0.40 F1)
'any' means context-independent — semantically ambiguous

Classifier C — Tone

Input: Utterance + context

Classes: neutral · positive · formal · negative · emotional

Why: Tone requires situational reading — Appraisal Theory framework

Macro F1: 0.61

Hardest: negative (0.41 F1)
Only 17 test examples for negative class

RESULTS — THE FULL ABLATION

STAGE	INPUT	ACCURACY	MACRO F1	W-F1	Δ MACRO F1
Stage 1	Utterance only	0.739	0.701	0.743	+ -
Stage 2	+ context + instruction	0.794	0.758	0.94	+ 0.057
Stage 3	+ oracle metadata	0.846	0.799	0.843	+ 0.041
Stage 4	+ predicted metadata	0.727	0.702	0.733	-0.097

S1→S2 +0.057: Adding social context meaningfully improves classification

S2→S3 +0.041: Perfect metadata is the oracle upper bound

S3→S4 −0.097: Error propagation from imperfect classifiers collapses the gain — Stage 4 ≈ Stage 1

KEY FINDINGS

FINDING 1: CONTEXT CARRIES REAL SIGNAL

Adding social context and instruction improves macro F1 by +0.057. The model cannot reliably classify pragmatic properties from text alone.

FINDING 2: METADATA HELPS — WHEN PERFECT

Providing exact register, power relation, and tone adds another +0.041. This is the oracle upper bound with ideal information.

FINDING 3: ERROR PROPAGATION IS THE BOTTLENECK

Predicted metadata (Stage 4) nearly negates the gain — dropping −0.097 back to Stage 1 level. Intermediate classifier quality is the critical factor.

FINDING 4: TONE NEEDS THEORETICAL GROUNDING

131 raw annotations are unlearnable. Mapping to Appraisal Theory (5 categories) makes the task tractable and academically defensible.

EXPERIMENT RESULTS

INTERMEDIATE CLASSIFIERS

CLASSIFIER	TASK	INPUT	CLASSES	MACRO F1
A	Register	Utterance	4	0.70
B	Power Relation	Utterance + Context	4	0.60
C	Tone	Utterance + Context	5	0.61

POLITENESS CLASSIFICATION - ABLATION

STAGE	INPUT	ACCURACY	MACRO F1	W-F1	TEST LOSS
Stage 1	Utterance only	0.739	0.701	0.743	1.043
Stage 2	+ Context + Instruction	0.794	0.758	0.794	0.804
Stage 3	+ Oracle Metadata	0.846	0.799	0.843	0.789
Stage 4	+ Predicted Metadata (Pipeline)	0.727	0.702	0.733	0.812

NEXT WEEK — FINAL SUBMISSION PLAN

Zero-Shot Baseline (Stage 0)

- Prompt GPT-4 and Gemini to classify politeness with no fine-tuning — just a description of the task and the utterance.
- Expected result: Our fine-tuned Stage 2 and Stage 3 models should outperform zero-shot LLMs on this specific Burmese pragmatics task. Burmese is under-represented in general LLM training, and pragmatic classification requires domain-specific signal that prompting alone cannot supply

Improve Intermediate Classifiers

- Target: push Classifiers A/B/C above 0.80 macro F1 to narrow the Stage 3→Stage 4 gap.
- Directions: (A) add context to register input; (B) merge 'any' in power relation; (C) refine tone mapping boundary.
- Expected: Stage 4 macro F1 rises above 0.702.

Error Analysis

- Examine specific misclassified Burmese examples — where does the model fail and why?
- Focus on Stage 3 vs Stage 4 disagreements: cases where oracle metadata leads to a correct prediction but predicted metadata leads to a wrong one. These examples directly illustrate the error propagation problem.

ALL MODELS ARE LIVE ON HUGGINGFACE

TWO INTERACTIVE DEMO SPACES

Demo v2 (Stage 4 pipeline):	annasus10/burmese-pragmatics-demo-v2
Stage 3 model (Best):	annasus10/xlmr-burmese-pragmatics-stage3-v2
Dataset:	freococo/burmese-contextual-pragmatics

THANK YOU!

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PROGRESS PRESENTATION